

DS 120 S25 Discussion 1 Solutions

DS 120 Staff

September 2025

Problem 1: Determine which of the following are considered mathematical statements. If applicable, determine their truth values.

(i) $2 + 2 = 4$

Solution: This is a true mathematical statement.

(ii) $2 + 2 = 5$

Solution: This is a false mathematical statement.

(iii) $2 + 2$

Solution: This is not a mathematical statement.

(iv) $x^2 = 2$

Solution: This is not a mathematical statement.

(v) $\exists x \in \mathbb{R} : x > 3$

Solution: This is a true mathematical statement.

(vi) $\forall x \in \mathbb{R} : x > -1$

Solution: This is a true mathematical statement.

(vii) 1,000,000 is a large number.

Solution: This is not a mathematical statement.

Problem 2: Translate the following into plain English and determine their truth values.

(i) $\exists x \in \mathbb{Z} : x < 0$

Solution: There exists a negative integer. This is true.

(ii) $\exists x \in \mathbb{N} : x < 0$

Solution: There exists a negative natural number. This is false.

(iii) $\forall x \in \mathbb{N}, \exists y \in \mathbb{R} : y > x$

Solution: For all natural numbers, x , there exists a real number that is greater than x . This is true.

Problem 3: Rewrite the following expressions using mathematical symbols.

(i) All natural numbers are greater than -1 .

Solution: $\forall x \in \mathbb{N}, x > -1$

(ii) π is not an integer.

Solution: $\pi \notin \mathbb{Z}$

(iii) A natural number multiplied by two is a natural number.

Solution: $\forall x \in \mathbb{N}, 2x \in \mathbb{N}$

(iv) A rational number can be written as the quotient of two integers with a nonzero denominator.

Solution: $\forall x \in \mathbb{Q}, \exists p, q \in \mathbb{Z} : q \neq 0 \wedge x = \frac{p}{q}$

(v) Between every two distinct real numbers, there is a rational number.

Solution: $\forall x, y \in \mathbb{R} : x < y, \exists q \in \mathbb{Q} : x < q < y$

Problem 4: Determine the following truth statements.

(i) $(2^3 = 9) \vee (3^2 = 9)$

Solution: True. This is the logical “or” operator. We only need one of the two to be correct for the entire statement to be true. Because $3^2 = 9$, this is true.

$$(ii) (2 + 2 = 5) \vee (2 * 3 = 5)$$

Solution: False, neither of these are true.

$$(iii) (\sqrt{4} \in \mathbb{N}) \wedge (\sqrt{2} \in \mathbb{Q})$$

Solution: False. This is the logical “and” operator. In other words, we need both to be true for the entire statement to be true. While 2 is a natural number, $\sqrt{2}$ is not rational.

$$(iv) (\sqrt{4} \in \mathbb{N}) \wedge (\sqrt{4} \in \mathbb{Z}) \wedge (\sqrt{4} \in \mathbb{Q}) \wedge (\sqrt{4} \in \mathbb{R})$$

Solution: This is true. 2 is a natural number, integer, rational number, and a real number.

$$(v) x < 2$$

Solution: This is a variable proposition because the truth value depends on the value of x . Therefore, it does not have a truth value.

$$(vi) (\{1, 2, 3, 4, 5, 6\} \subseteq \mathbb{N}) \wedge (\mathbb{N} \subseteq \mathbb{R})$$

Solution: True. This deals with subsets. $\{1, 2, 3, 4, 5, 6\}$ are all contained within $\mathbb{N}$ and all the natural numbers are contained within the set of real numbers.